ROC Analysis

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Introduction

The receiver-operating-characteristic (ROC) curve traces sensitivity against $1 - $ specificity across all possible decision thresholds of a continuous-valued diagnostic test or risk score. Where a single sensitivity-specificity pair characterises performance at one cut-off, the ROC curve summarises performance across every cut-off and therefore captures the discriminative ability of the underlying continuous measurement independently of any chosen threshold. The area under the curve (AUC) reduces this curve to a single number with a clean probabilistic interpretation: it equals the probability that a randomly chosen diseased case has a higher biomarker value than a randomly chosen non-diseased case. AUC values of 0.5 indicate chance-level discrimination and 1.0 indicates perfect separation; values between 0.7 and 0.9 are typical for clinically useful biomarkers.

Prerequisites

A working understanding of sensitivity and specificity, the role of decision thresholds in diagnostic-test performance, and the trade-off between true and false positive rates that the ROC curve makes explicit.

Theory

For a continuous biomarker X and binary disease status D, the ROC curve is the parametric plot

$$\mathrm{ROC}(c) = \big(1 - F_0(c),\, 1 - F_1(c)\big) \quad \text{for all } c,$$

where F_0 and F_1 are the CDFs of X in non-diseased and diseased populations, respectively. The AUC has the equivalent representation

$$\mathrm{AUC} = P(X_1 > X_0),$$

with X_1 a random observation from the diseased and X_0 from the non-diseased population — the probability of correct ranking. Confidence intervals on AUC are typically computed via the DeLong non-parametric method or by bootstrap; bootstrap is also the standard approach for inference on the curve itself.

Assumptions

Test values are measured on a continuous scale (or at least an ordinal scale with many levels), disease status is correctly classified by a reliable gold-standard reference, and observations are independent. The interpretation as a discrimination measure is independent of disease prevalence — a feature of AUC that distinguishes it from predictive values.

R Implementation

```r
library(pROC)

set.seed(2026)
```

```r
n <- 200
disease <- factor(sample(c(0, 1), n, replace = TRUE, prob = c(0.6, 0.4)))
biomarker <- rnorm(n, mean = ifelse(disease == 1, 1.0, 0), sd = 1)

roc_obj <- roc(response = disease, predictor = biomarker,
               levels = c("0", "1"), direction = "<")
auc(roc_obj)
ci.auc(roc_obj, method = "delong")

plot(roc_obj, col = "#2A9D8F", lwd = 2, legacy.axes = TRUE,
     main = "ROC curve for biomarker")
abline(0, 1, lty = 2, col = "grey60")
```

Output & Results

`roc()` constructs the ROC object; `auc()` returns the area under the curve and `ci.auc()` provides the DeLong or bootstrap confidence interval. The standard plot shows sensitivity on the vertical axis and $1 - $ specificity on the horizontal axis, with the chance-diagonal as a reference line. Points along the curve correspond to sensitivity-specificity trade-offs at different cut-off values.

Interpretation

A reporting sentence: "The biomarker showed good discrimination for the binary disease outcome with AUC 0.77 (95 % CI 0.70 to 0.84, DeLong method); this exceeds the conventional 'fair' threshold of 0.70 but falls short of 'excellent' (≥ 0.90). At the Youden-optimal cut-off (biomarker ≥ 0.42), sensitivity was 73 % and specificity 70 %; alternative cut-offs prioritising specificity (e.g., ≥ 1.0, sensitivity 51 %, specificity 87 %) may be preferred for screening applications." Always report both AUC and at least one operating point.

Practical Tips

- Report AUC with a 95 % confidence interval — DeLong's non-parametric method is the standard for paired comparisons and bootstrap is preferable for very small or imbalanced samples; AUC without uncertainty bounds is uninterpretable.
- Test AUC against 0.5 (chance) using `pROC::roc.test()`; compare two AUCs from the same cases using a paired DeLong test, which respects the correlation induced by shared subjects.
- Partial AUC over a clinically relevant region — for example, AUC restricted to specificity above 0.9 in a screening context — is often more informative than full AUC, because clinical use rarely spans the full operating range.
- AUC is threshold-independent; combine it with a calibration analysis (Hosmer-Lemeshow, calibration intercept and slope, calibration plot) or a decision-curve analysis when threshold-dependent decisions matter for the clinical context.
- For heavily imbalanced data (rare disease, screening contexts), precision-recall AUC is often more informative than ROC AUC because the ROC can look deceptively good when most subjects are non-diseased; PRAUC focuses on the positive class.
- When comparing biomarkers, include the increment in AUC (Δ AUC), the integrated discrimination index (IDI), and the net reclassification improvement (NRI); each captures a different facet of incremental performance.

R Packages Used

`pROC` for canonical ROC analysis with DeLong and bootstrap CIs, partial AUC, and paired comparisons; `ROCR` for an alternative interface with comprehensive performance-measure support; `PRROC` for precision-recall AUC and area-under-the-PR-curve analyses; `cutpointr` for principled threshold selection; `rms::lrm()` for AUC reporting integrated with logistic-regression model evaluation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale